

When the Fundamental Is Missing: An Experimental Investigation of Virtual Pitch

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Abstract—This report investigates the psychoacoustic phenomenon of the “missing fundamental,” where the auditory system perceives a pitch corresponding to a fundamental frequency (f_0) even in its spectral absence. Using a custom MATLAB analysis tool, six controlled test cases ranging from isolated complex tones to randomized melodic contexts were evaluated to stress-test the limits of pitch perception theories. The results demonstrate that while temporal phase-locking explains high-frequency limits, pattern recognition mechanisms are essential for establishing pitch in inharmonic and complex spectral contexts.

Index Terms—Virtual pitch, missing fundamental, pitch perception, psychoacoustics, harmonic series, additive synthesis.

I. INTRODUCTION

Pitch perception is a fundamental attribute of auditory sensation. Classically, the Place Theory posits that pitch is determined by the point of maximum excitation along the basilar membrane, a concept extensively discussed in standard acoustics literature [1]. However, the phenomenon of the Missing Fundamental (or Virtual Pitch) presents a significant paradox: the human auditory system perceives a pitch corresponding to a fundamental frequency (f_0) even when spectral energy at that frequency is physically absent.

As noted by Howard and Angus [1], this phenomenon is not merely a laboratory curiosity but an essential component of everyday hearing. For instance, it allows listeners to identify the pitch of a male voice over a telephone line, where the limited bandwidth (approx. 300 Hz – 3.4 kHz) removes the fundamental frequency, yet the brain reconstructs the correct pitch based on the remaining upper harmonics.

This report presents the design and evaluation of a MATLAB-based interactive tool developed to demonstrate and analyse the limits of this psychoacoustic illusion. Through six controlled test cases, ranging from basic complex tones to melodies with randomized spectral profiles, this investigation explores how variables such as harmonic content, frequency limits, and inharmonicity affect the robustness of pitch perception. The results are critically discussed in relation to the competing theories of Temporal Processing [2] and Pattern Recognition [3].

II. BACKGROUND THEORY

Pitch perception is traditionally explained by the Place Theory [1], which links pitch to the location of cochlear stimulation. Nevertheless, this model fails to explain the “missing fundamental” phenomenon, where a pitch of frequency f_0 is perceived despite the absence of spectral energy at that

frequency, provided that upper harmonics ($nf_0, (n+1)f_0 \dots$) are present.

Two primary theoretical frameworks compete to explain this mechanism:

A. Temporal Theory (Periodicity Pitch)

This model suggests that the auditory system analyses the time-domain waveform envelope. Even in the absence of the fundamental component, the superposition of higher harmonics creates a beat periodicity equal to $1/f_0$. Thurlow and Hartman [2] investigate this effect, highlighting that neural “phase-locking” allows the brain to track these temporal intervals. However, this mechanism is biologically limited by the refractory period of auditory neurons, typically failing at frequencies above 4–5 kHz, a range known as the “existence region” of virtual pitch.

B. Pattern Recognition Theory (Spectral Pitch)

Alternatively, central processing models propose that the brain resolves individual partials in the cochlea and performs a template-matching operation to find the best-fitting harmonic series [3]. Zwicker and Fastl describe this as “residue pitch,” noting that the effect is most robust when at least three consecutive harmonics are spectrally resolved.

Recent studies by Chialvo [4] further suggest that non-linear neural dynamics may play a role in reconstructing the missing information through sub-threshold neural interactions, allowing the auditory system to recover periodicity from incomplete signals.

III. METHODOLOGY AND IMPLEMENTATION

To evaluate the limits of the theoretical models discussed, a MATLAB-based analysis tool was developed using additive synthesis. Unlike subtractive methods, this approach allows for precise control over the amplitude (A_k), frequency (f_k), and phase (ϕ_k) of each individual partial, enabling the isolation of specific psychoacoustic cues.

A. Signal Generation Framework

The core synthesis logic is encapsulated in a custom function (`generateComplexWave.m`) that constructs complex tones by summing sinusoidal components according to equation (1):

$$y(t) = \sum_{k=1}^N A_k \sin(2\pi f_k t + \phi_k) \quad (1)$$

Where N is the number of harmonics and f_0 is the fundamental frequency. The amplitude distribution A_k is determined by the specific waveform profile:

- **Square Wave:** Simulates odd harmonics only ($f_k = kf_0$ for $k = 1, 3, 5 \dots$) with amplitudes $A_k = 1/k$.
- **Sawtooth Wave:** Simulates a full harmonic series ($k = 1, 2, 3 \dots$) with $A_k = 1/k$. This profile provides the strongest periodicity cues for virtual pitch retrieval.

B. Experimental Case Design

Six test cases were implemented to stress-test the auditory system's pitch extraction mechanisms:

- 1) *Harmonic Density (Cases 1-2):* A comparison between square and sawtooth waves to determine if the spectral gaps in odd-harmonic signals affect the perceived pitch strength.
- 2) *The Existence Region (Case 3):* A high-frequency test case where f_0 is raised to 6 kHz. This tests the upper limits of the Temporal Theory, as phase-locking typically degrades above 4–5 kHz [2].
- 3) *Spectral Dominance (Case 4):* A custom algorithm ("F2 Dominant") was implemented where the fundamental is removed ($A_1 = 0$) and higher harmonics ($k > 2$) are attenuated by an additional constant factor of 0.1. This isolates the second harmonic ($2f_0$) as the spectral peak, testing whether pitch perception follows the energy maximum or the harmonic pattern.

C. Inharmonicity Simulation

To challenge the Pattern Recognition model [3], Case 5 modifies the synthesis equation to introduce a fixed frequency shift Δf to every harmonic, as shown in (2):

$$f_k = k \cdot f_0 + \Delta f \quad (2)$$

With $\Delta f = 50$ Hz, the components (250, 450, 650... Hz) no longer form an integer ratio series, theoretically disrupting the "residue pitch" formation.

D. Contextual Robustness

Finally, Case 6 generates a melody ("The Imperial March") where the spectral composition of each note is randomized using a uniform distribution. A custom function (`generateStressMelody.m`) assigns one of five spectral profiles—ranging from full spectrum to missing fundamental—to each note event. This evaluates the brain's ability to maintain melodic continuity despite inconsistent spectral cues.

E. Analysis Visualization

The signal output is analysed using Short-Time Fourier Transform (STFT) spectrograms with a Hanning window (length 2048, overlap 50%). This allows for a direct visual comparison between the physical energy distribution (frequency domain) and the perceived auditory sensation.

IV. RESULTS AND DISCUSSION

The developed MATLAB tool was utilised to evaluate the test cases defined. The resulting audio signals were analysed both aurally and visually using the generated STFT spectrograms.

A. Case 1: Harmonic Density (Square Wave)

Case 1 utilised a Square Wave with a fundamental frequency of $f_0 = 200$ Hz. This waveform contains only odd harmonics (600, 1000, 1400... Hz), as illustrated in Fig. 1. Despite the physical absence of energy at 200 Hz, the pitch was correctly identified at the fundamental frequency. However, the auditory sensation was somewhat "hollow" due to the sparse spectral cues.

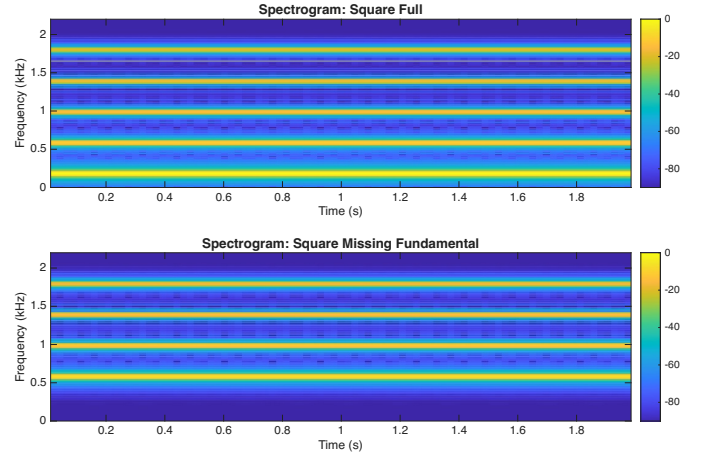


Fig. 1. Case 1 ($f_0 = 200$ Hz, Square Wave). Note the gaps in the spectrum (odd harmonics only) which creates a weaker virtual pitch cue.

B. Case 2: Harmonic Density (Sawtooth Wave)

Case 2 utilised a Sawtooth Wave at $f_0 = 300$ Hz. Unlike the previous case, Fig. 2 shows that this waveform contains all integer harmonics (600, 900, 1200... Hz). The perceived pitch at 300 Hz was significantly more robust and unambiguous than in Case 1. This aligns with the Pattern Recognition Theory [3], as the complete harmonic series provides redundant spectral data points, allowing the central processor to match the harmonic template more effectively.

C. Case 3: The Existence Region Limit

In Case 3, the fundamental frequency was raised to $f_0 = 5000$ Hz (5 kHz). At this frequency, the virtual pitch percept deteriorated. As seen in Fig. 3, the auditory percept did not correspond to a 5 kHz missing fundamental, but rather to the distinct high-pitched tones of the individual harmonics (10 kHz, 15 kHz). This result supports the biological constraint proposed by the Temporal Theory [2]: neural phase-locking degrades significantly above 4–5 kHz, preventing the reconstruction of the fundamental.

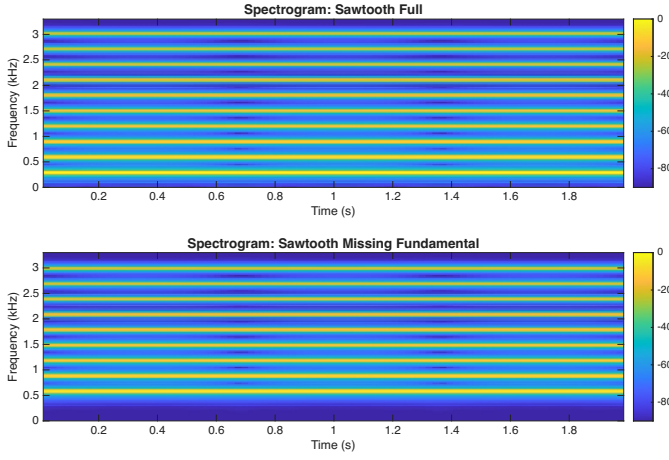


Fig. 2. Case 2 ($f_0 = 300$ Hz, Sawtooth Wave). The presence of all integer harmonics reinforces the “missing fundamental” perception.

D. Case 4: Spectral Dominance

Case 4 was evaluated at $f_0 = 400$ Hz using the “F2 Dominant” algorithm. The fundamental (400 Hz) was removed, and higher harmonics ($k > 2$) were strongly attenuated (by 90%) isolating the second harmonic (800 Hz) as the dominant spectral peak to contain the majority of the spectral energy, as highlighted in Fig. 4. Crucially, the perceived pitch remained at 400 Hz, not the spectrally dominant 800 Hz. This confirms that pitch extraction is a derived property based on harmonic relationships, not merely a detection of the maximum energy peak [3].

E. Case 5: Inharmonicity

Case 5 applied a fundamental frequency of $f_0 = 350$ Hz with a non-integer frequency shift of $\Delta f = 50$ Hz applied to the partials. Fig. 5 displays how this disruption of the integer ratios caused the unified pitch percept to deteriorate into an ambiguous, bell-like sound. The brain struggled to fit the inharmonic components to a single harmonic template of

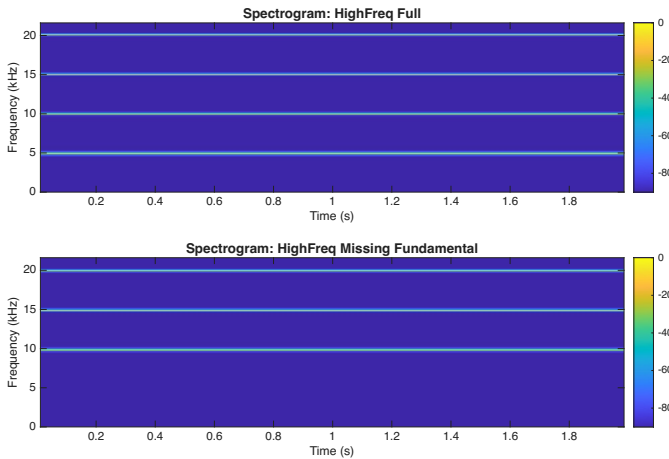


Fig. 3. Case 3 ($f_0 = 5000$ Hz). The harmonics are widely spaced and exceed the phase-locking limit. No virtual pitch is perceived.

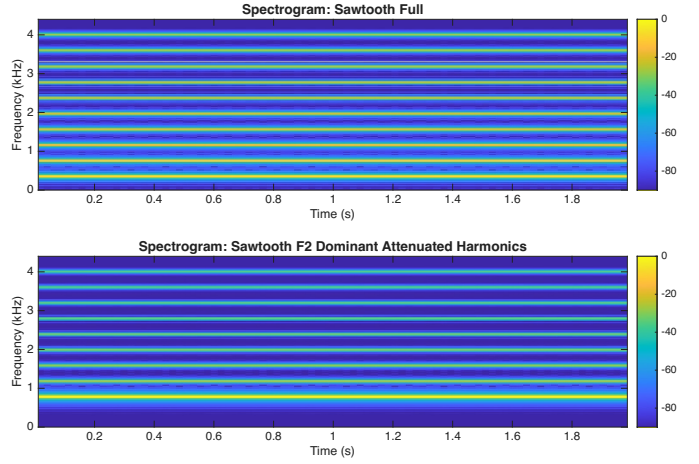


Fig. 4. Case 4 ($f_0 = 400$ Hz, F2 Dominant). Despite the energy peak at 800 Hz ($2f_0$), the pitch is still perceived at the missing fundamental of 400 Hz.

350 Hz, validating the strict spectral requirements proposed by Zwicker and Fastl [3].

F. Contextual Robustness (Melody)

Finally, Case 6, the randomized melody stress-test (“The Imperial March”) demonstrated that even with rapidly changing spectral profiles (randomly switching between full spectrum, missing fundamental, F2 dominant, Extreme (No F1-F5) and Inharmonicity, as detailed in Table I), the melodic contour remains intelligible. The spectrogram in Fig. 6 visualizes this highly irregular spectral content. This suggests that top-down cognitive processes integrate pitch cues across time, allowing the auditory system to maintain perceptual coherence despite frame-by-frame spectral inconsistencies.

V. CONCLUSION

This investigation successfully demonstrated that pitch perception is a complex, non-linear process that extends beyond simple spectral analysis. The results obtained from the

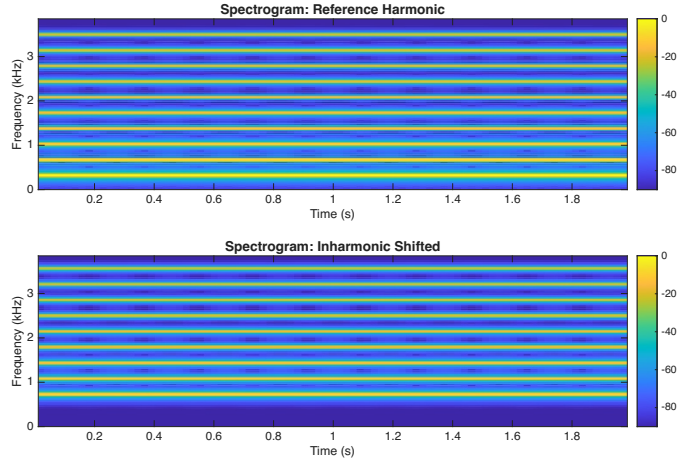


Fig. 5. Case 5 ($f_0 = 350$ Hz, Inharmonic). The spectral shift disrupts the integer multiples, creating pitch ambiguity.

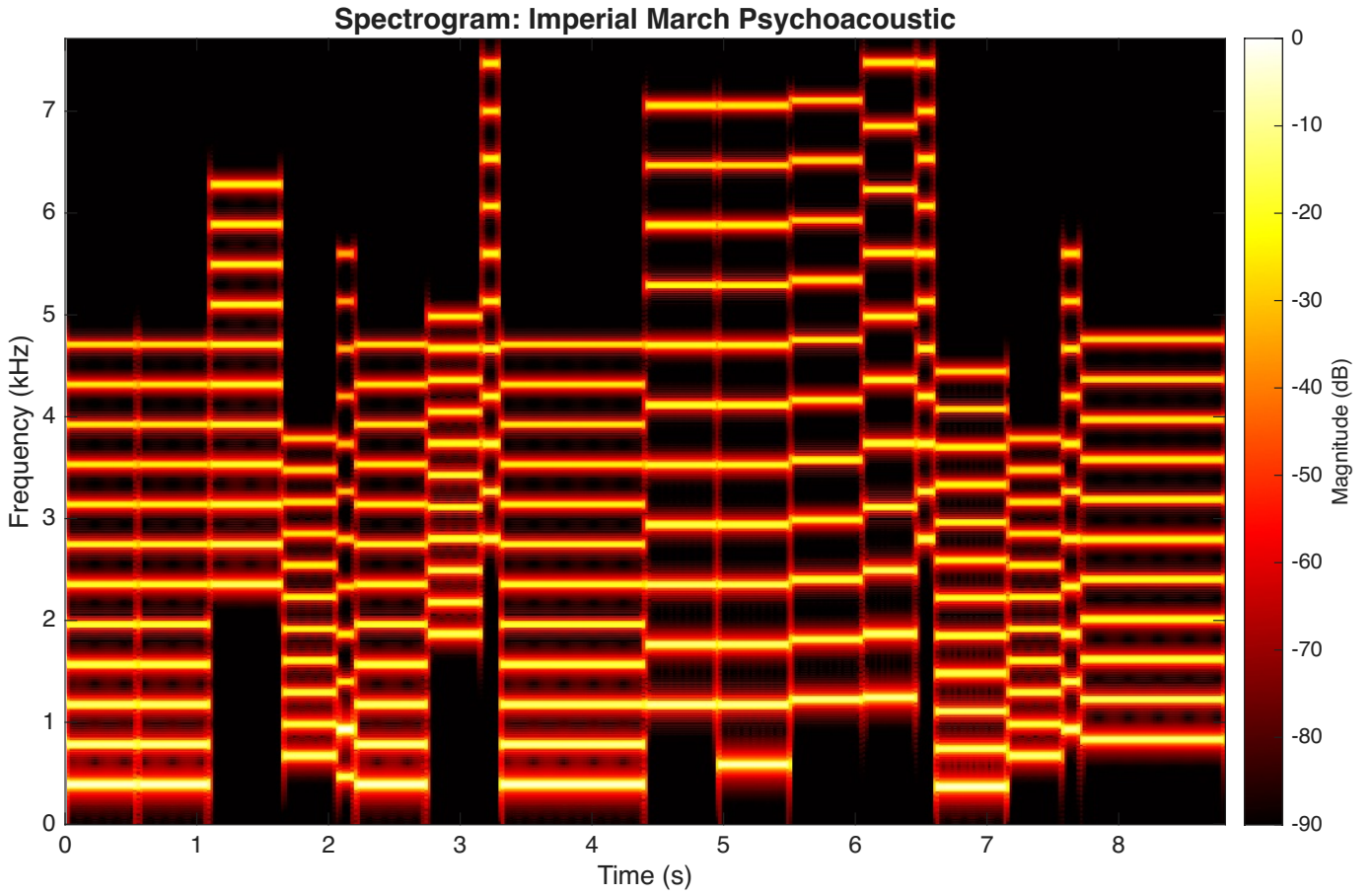


Fig. 6. Spectrogram of the randomized melody test. The melodic contour remains recognisable despite rapidly varying spectral content across notes

TABLE I
RANDOMIZED MELODY SEQUENCE PARAMETERS

#	Note	f_0	Cond.	#	Note	f_0	Cond.
1	G4	392	Full	10	D5	587	Miss F1
2	G4	392	Full	11	D5	587	Full
3	G4	392	Extr*	12	D5	587	Inharm
4	Eb4	311	Inharm	13	Eb5	622	Miss F1
5	Bb4	466	F2 Dom	14	Bb4	466	Extr*
6	G4	392	Full	15	Gb4	370	Full
7	Eb4	311	Extr*	16	Eb4	311	Inharm
8	Bb4	466	Extr*	17	Bb4	466	Miss F1
9	G4	392	Full	18	G4	392	Inharm

Key: **Extr***: Extreme (No f_1 - f_5), **Inharm**: Inharmonic, **Miss F1**: Missing Fundamental, **F2 Dom**: 2nd Harmonic Dominant.

robustness of the effect in the “F2 Dominant” and randomized melody cases highlights the auditory system’s ability to reconstruct reality from incomplete data, a mechanism essential for speech intelligibility in noise and telecommunications [1].

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MATLAB simulation challenge the classical Place Theory, showing that the fundamental frequency is not required for pitch identification.

The experiments confirm that while Temporal Theory explains the frequency limits of the phenomenon (failing > 5 kHz), the Pattern Recognition Theory better accounts for the brain’s requirement for harmonic integer relationships. The